Food Production Optimization

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***Abstract*—Food production is the process by which land, water, labor, and other inputs are coordinated to generate the nutrients that sustain human populations – and optimizing it at scale is one of the defining resource allocation challenges of the coming decades, as rising demand, climate volatility, and sustainability constraints simultaneously tighten.**

We study a multi-objective crop allocation problem balancing nutritional value, environmental sustainability, and affordability for 27 distinct crops. We thereby evaluate the engineering and mathematical trade-offs of mapping large-scale structured tasks to quantum annealing hardware. The problem is modeled as a Binary Plot (BP) formulation: a grid of fixed-area plots, each assigned to exactly one crop, expressed as a Constrained Quadratic Model (CQM). We benchmark our approach against the classical Gurobi solver using problems with 10 to 1,000 land patches and run two experiments: (1) a comparative analysis utilizing D-Wave's hybrid solvers, and (2) an evaluation of pure Quantum Processing Unit (QPU) execution using custom decomposition algorithms. Our results demonstrate that migrating the discrete BP formulation to quantum hardware exposes severe classical embedding bottlenecks: although QPU execution time scales linearly, the classical preprocessing overhead required to embed densely connected penalty models constitutes 76–86% of the total runtime. We conclude that for structured allocation tasks, overcoming this classical embedding bottleneck—rather than accelerating QPU computation—remains the primary frontier for realizing quantum utility.

***Index Terms*—QUBO, Annealing, SDGs, D-Wave**

I. Introduction

Global agriculture must simultaneously meet the rising global caloric demand, while providing food of sufficient nutritional quality, environmental sustainability, and economic accessibility—challenges at the core of the UN Sustainable Development Goals [1], [2]. Conventional planning focuses on yield maximisation, yet modern frameworks must jointly optimise nutrition, ecological footprint, and affordability [3]–[5]. When farmland is partitioned into discrete plots, each assigned to exactly one crop, the resulting combinatorial allocation problem maps naturally to a Binary Plot (BP) formulation expressible as a Constrained Quadratic Model [6], [7].

Quantum annealing (QA) encodes combinatorial objectives as physical Ising energy landscapes, exploiting quantum tunneling to traverse barriers that trap classical heuristics [8], [9]. The approach has been applied to graph colouring [10], scheduling [11], and portfolio optimisation [12], among others, and is reviewed comprehensively by Yarkoni et al. [13]. Mapping structured allocation problems onto current D-Wave hardware [14], [15], however, exposes a fundamental engineering bottleneck: enforcing one-hot constraints via quadratic penalty terms [16], [17] introduces $O(|\mathcal{P}|\times|\mathcal{C}|^2)$ couplings, producing Binary Quadratic Models (BQMs) too densely connected for the Pegasus topology [18]. Classical minor-embedding algorithms must then map logical variables to chains of physical qubits [19], [20], a step that incurs quadratic space overhead and polynomial time-to-solution penalties [21]–[23]. Penalty encoding undermines the structured constraint information that classical branch-and-bound exploits [7], rendering solvers like Gurobi ineffective on the penalised form—yet QA's energy-minimisation paradigm is not similarly affected.

This paper presents a systematic engineering study of the BP crop allocation problem on D-Wave Advantage hardware. We benchmark hybrid and direct-QPU workflows against Gurobi across 10–1,000-plot instances and evaluate eight decomposition strategies that partition large problems into QPU-compatible subproblems [24], [25]. Our principal finding is that QPU annealing scales favourably and remains fast at every tested scale, while classical embedding accounts for 76–86% of end-to-end runtime—establishing embedding, not sampling, as the binding bottleneck.

II. Problem Formulation

- **Sets and indices**
  - $\mathcal{P}$: set of plots, indexed by p, each with fixed area L_p (ha)
  - $\mathcal{C}$: set of crops, indexed by c, with $|\mathcal{C}| = 27$
  - $\mathcal{G}$: set of food groups, indexed by g, with $|\mathcal{G}| = 5$
  - $\mathcal{C}_g \subseteq \mathcal{C}$: subset of crops belonging to food group g
- **Decision variables**

- $Y_{p,c} \in \{0,1\}$: equals 1 if plot p is assigned to crop c, 0 otherwise
- $U_c \in \{0,1\}$: equals 1 if crop c is planted on at least one plot

- **Parameters (per crop c)**
  - $n_c^{\min/\max}$ Minimum (maximum) number of plots assigned to a certain crop
  - $v_{nv,c}$: nutritional value, capturing the crop's contribution to diet quality through vitamins, minerals, and fiber
  - $v_{nd,c}$: nutrient density, capturing how much nutritional content the crop provides relative to calories and weight
  - $v_{ei,c}$: environmental impact, capturing the crop's climate-change burden
  - $v_{af,c}$: affordability, capturing how inexpensive the crop is per unit of food output
  - $v_{su,c}$: sustainability, capturing the crop's overall life-cycle environmental burden across all available LCA categories and stages, excluding the ones included in $v_{ei,c}$

Dataset. We use GAIN's agricultural data from Bangladesh and Indonesia covering 27 crops across 5 food groups (animal-source foods, fruits, legumes, starchy staples, vegetables). Each crop c receives a composite benefit score

$$B_c = w_{nv}v_{nv,c} + w_{nd}v_{nd,c} - w_{ei}v_{ei,c} + w_{af}v_{af,c} + w_{su}v_{su,c}, \tag{1}$$

integrating nutritional value, nutrient density, environmental impact (negatively weighted), affordability, and sustainability, with $\sum_i |w_i|=1$ ($w_{nv}=0.25$, $w_{nd}=0.20$, $w_{ei}=0.20$, $w_{af}=0.20$, $w_{su}=0.15$, arbitrarily chosen).

Binary Plot (BP) model. Let $\mathcal{P}$ be the set of plots, $\mathcal{C}$ the set of crops, $\mathcal{G}$ the set of food groups, and $U_c \in \{0,1\}$ a global indicator for whether crop c is planted anywhere. Each plot p of area L_p is assigned entirely to one crop via $Y_{p,c} \in \{0,1\}$. The objective maximises area-weighted benefit:

$$\max\ Z = \frac{1}{A_{\text{total}}} \sum_{p\in\mathcal{P}} \sum_{c\in\mathcal{C}} L_p \cdot B_c \cdot Y_{p,c}, \tag{2}$$

where $A_{\text{total}}=\sum_p L_p$. Constraints enforce one-hot plot assignment ($\sum_c Y_{p,c} \le 1$), mandatory food-group coverage ($\sum_{c\in\mathcal{C}_g} U_c \ge 1$ for each $g \in \mathcal{G}$), linkage ($Y_{p,c} \le U_c$), and per-crop area bounds ($n_c^{\min} \cdot U_c \le \sum_p Y_{p,c} \le n_c^{\max} \cdot U_c$). The model contains $|\mathcal{P}|\times|\mathcal{C}|+|\mathcal{C}|$ binary variables and no continuous variables, making it natively compatible with CQM and QUBO formulations.

Computational complexity. The decision version of BP—does there exist an assignment with $Z \ge k$?—is NP-complete, which we establish by showing its constraint structure requires at least 3-SAT and cannot be reduced to 2-SAT. Consider the CNF encoding of the two main constraint families. The *at-most-one* part of the one-hot constraint ($\neg Y_{p,c} \vee \neg Y_{p,c'}$ for all $c \neq c'$) produces only pairwise 2-literal clauses, which fall within the 2-SAT fragment and are solvable in polynomial time [26]. However, the *food-group coverage* constraint requires that at least one crop from each group is planted somewhere:

$$\bigvee_{c\in\mathcal{C}_g} U_c \quad \forall\, g \in \mathcal{G}. \tag{3}$$

Each group in our dataset contains $|\mathcal{C}_g| \ge 2$ crops (minimum 2 for starchy staples, up to 9 for fruits), so every coverage clause has at least 2 literals; for all groups with $|\mathcal{C}_g| \ge 3$ the clause length exceeds 2 and cannot be decomposed into 2-clauses without introducing auxiliary variables. Since these long clauses appear alongside the binary objective, the full problem does not reduce to 2-SAT and is NP-hard in general [27], [28]. This places BP firmly in the class of problems for which heuristic and quantum approaches are practically motivated.

Penalty encoding for QPU. For direct QPU execution, one-hot constraints are absorbed as quadratic penalties:

$$H_{\text{penalty}} = \lambda \sum_{p\in\mathcal{P}} \left(\sum_{c\in\mathcal{C}} Y_{p,c} - 1 \right)^2, \tag{4}$$

introducing $O(|\mathcal{P}|\times|\mathcal{C}|^2)$ quadratic couplings. The full BQM energy is $E(\mathbf{x})=\sum_i h_i x_i + \sum_{i<j} J_{ij} x_i x_j$, with linear biases $h_{(p,c)}=-L_p B_c/A_{\text{total}}$ and couplings $J_{(p,c_1),(p,c_2)}=2\lambda$ for $c_1 \neq c_2$. This QUBO form is mandatory for annealing but substantially increases problem density relative to the native formulation, degrading classical solver performance [16], [17], [29].

III. Methodology

Quantum hardware. All experiments ran on D-Wave Advantage (Pegasus P16, 5,760 qubits, avg. connectivity 15, native cliques of 15–20 qubits) via the Leap cloud platform [14], [15], [30]. Recent characterisation confirms coherent quantum dynamics in programmable Ising chains on this processor [31]. For direct QPU runs: 100–500 reads per subproblem, 20 μs annealing time. Observed chain break rates remained below 2% across all instances, placing embeddings in the high-fidelity regime [20], [23], [32], [33]. The Pegasus topology's native cliques of 15–20 qubits are central: 27-variable plot subproblems embed with average chain length $\le$1.2 [18], [34], yielding predicted break rates of $\sim$1.8% ($1-(1-\varepsilon)^k$ with $\varepsilon \approx 1.5\%$, $k=1.2$), consistent with observation. For hybrid runs, LeapHybridCQMSolver and LeapHybridBQMSolver used default parameters.

Classical baseline. Gurobi 12.0.1 on Intel Core i7-12700H (14 cores, 16 GB RAM); 300 s time limit; MIPGap 1%; MIP Focus mode 1 (feasibility priority); multi-threading enabled [7]. Reported times are wall-clock values including all preprocessing.

Decomposition strategies. Direct QPU embedding is infeasible above $\sim$180 variables due to Pegasus connectivity limits [18], [21], [23]. We therefore evaluate eight decomposition strategies that partition large instances into QPU-compatible subproblems [24], [25], spanning a range of partitioning philosophies: multilevel graph coarsening [35], [36], Louvain community detection [37], spectral clustering [38], [39], plot-level decomposition, grid-based decomposition, and

TABLE I
GUROBI ON BP: CQM (EXPLICIT CONSTRAINTS) VS. PENALTY-ENCODED BQM. CQM FINDS $Z^\star$ AT ALL SCALES IN UNDER 1 S; BQM CONSISTENTLY TIMES OUT AT 300 S AND RETURNS $Z{=}0$ ABOVE 500 PATCHES.

Patches	CQM Time (s)	CQM Obj. ($Z^\star$)	BQM Time (s)	BQM Obj.	BQM Status
10	0.007	0.409	300.2	0.284	Timeout
15	0.007	0.413	301.5	0.305	Timeout
25	0.007	0.388	301.7	0.215	Timeout
50	0.011	0.242	301.4	0.238	Timeout
100	0.028	0.429	302.1	0.254	Timeout
200	0.048	0.430	302.8	0.018	Timeout
500	0.453	0.430	360.7	0.000	Timeout
1,000	0.266	0.430	769.2	0.000	Timeout

TABLE II
D-WAVE HYBRID SOLVER PERFORMANCE (STUDY 1). BOTH CQM AND BQM SOLVERS RETURN INFEASIBLE SOLUTIONS (MARKED *) AT EVERY TESTED SCALE; SEE TABLE III.

	CQM Hybrid			BQM Hybrid		
Plots	Time (s)	QPU (ms)	Obj.	Time (s)	QPU (ms)	Obj.
10	5.38	70	0.743*	3.00	52	0.223*
50	5.29	70	0.347*	4.36	104	0.194*
100	5.35	70	0.549*	18.3	310	7.857*
500	5.67	35	3.426*	182	1,241	23.6*
1,000	11.04	35	3.757*	344	672	22.7*

TABLE III
HYBRID SOLVER INFEASIBILITY (STUDY 1): OBJECTIVE INFLATION (RAW − HEALED) AND ACTUAL ONE-HOT VIOLATION COUNT PER SOLVER. VIOLATION COUNTS ARE COMPUTED BY POST-PROCESSING: FOR EACH PATCH, WE COUNT CROPS WITH $Y_{p,c}{>}0.5$ AND FLAG PATCHES WHERE THIS EXCEEDS 1. THE BUILT-IN VALIDATOR COUNTER IS UNRELIABLE AT $n{\geq}50$ (REPORTS ZERO DESPITE LARGE INFLATION). ALL VIOLATIONS ARE ONE-HOT TYPE; NO FOOD-GROUP VIOLATIONS OCCUR IN STUDY 1.

Patches	CQM Infl.	CQM act.	BQM Infl.	BQM act.
10	0.60	7	0.12	9
15	0.44	12	0.04	10
25	0.31	2	0.31	11
50	0.26	10	0.18	2
100	0.43	18	7.70	100
200	3.06	200	10.93	200
500	3.23	500	23.40	500
1,000	3.56	1,000	22.51	1,000

an iterative master-subproblem approach analogous to classical domain decomposition [40], [41]. One variant—CQM-First PlotBased—partitions at the CQM constraint level before BQM conversion rather than after, which avoids penalty-term cross-contamination across partition boundaries [42], [43].

IV. RESULTS

A. CQM vs BQM Submission Format

When the BP problem is submitted to Leap's HybridCQMSolver and constraints are explicit, Gurobi processes it efficiently: under 0.27 s even at 1,000 plots (27,027 variables). Table I shows what happens when those same constraints are instead formulated via penalties (when the problem is submitted to the HybridBQMSolver). Gurobi consistently hits its 300 s timeout at all scales from 10 to 1,000 (reporting nominal "Optimal" status with objective values that deteriorate sharply—from 0.284 at 10 patches to exactly 0.0 at 500 and 1,000 patches—indicating the solver cannot make meaningful progress on the penalised form). Meanwhile, Gurobi on the CQM formulation solves all instances in well under 4 s, confirming that penalty encoding destroys the algebraic constraint structure that branch-and-bound exploits [7], [29].

B. Hybrid Solver Benchmark

Four solver profiles emerge:

1) **Gurobi (CQM):**Gurobi solves all BP instances optimally across 10–1,000 plots, reaching an objective of $Z^\star{\approx}0.430$ at 1,000 plots in 0.27 s (fig. 1, left panel).
2) **D-Wave CQM (Hybrid):** Near-constant $\sim$5.3 s runtime up to 500 patches, increasing to 11.0 s at 1,000 patches. QPU time constitutes only 0.3–1.3% of wall-clock time. The CQM solver returns infeasible solutions at *every* tested scale: healed-objective analysis (see section IV-D) reveals objective inflation (raw − healed) of 0.26–0.60 at $n{\leq}100$ and 3.1–3.6 at $n{\geq}200$, indicating systematic multi-crop over-assignment that the solver's own constraint validator fails to detect. The raw CQM gap reaches $|3.76-0.43|{\approx}3.3$ at 1,000 patches.
3) **D-Wave BQM (Hybrid):** Runtime grows from 3.0 s ($n{=}10$) to 344 s ($n{=}1{,}000$)—over two orders of magnitude—since BQM handles the penalty-encoded QUBO form where Gurobi fails entirely. The BQM solver also produces infeasible solutions at all scales: objective inflation grows from 0.04–0.31 at $n{\leq}25$ to 7.7 ($n{=}100$) and 22.5–23.4 ($n{\geq}500$), reflecting severe multi-crop over-assignment that escalates with problem size.
4) **Gurobi (BQM):** Consistently hits the 300 s timeout. Returned objectives degrade from 0.284 ($n{=}10$) to 0.000 ($n{\geq}500$), confirming that penalty encoding renders classical branch-and-bound ineffective.

Feasibility and healed gaps. Both D-Wave solvers return infeasible solutions at every tested scale, as revealed by healed-objective analysis (table III). Table III shows actual violation counts (patches with $Y_{p,c}{>}0.5$ for more than one crop, computed by post-processing). Figure 1 (middle panel) reports the *absolute solution gap* $|Z_{\text{solver}}-Z^\star|$ for both the raw solution and a post-hoc "healed" version, in which multi-crop over-assignments are repaired by retaining only the highest-benefit crop per patch (see section IV-D). The healed CQM gap is modest at small scales (e.g. $3.5{\times}10^{-3}$ at $n{=}50$) but grows to $\sim$0.1–0.2 at larger scales. The healed BQM gap remains above 0.1 across all scales. Gurobi BQM maintains a consistent gap of $\sim$0.05–0.33 (with no constraint violations—its solutions are feasible, merely suboptimal).

Figure 1 presents the full comparison.

C. QPU Decomposition and the Embedding Bottleneck

Table IV isolates QPU annealing time from classical embedding time for Multilevel(10) decomposition. QPU time follows $T_{\text{QPU}} \approx k \cdot n_{\text{part}} \cdot t_{\text{anneal}}$, where $k{=}1$–3 coordination

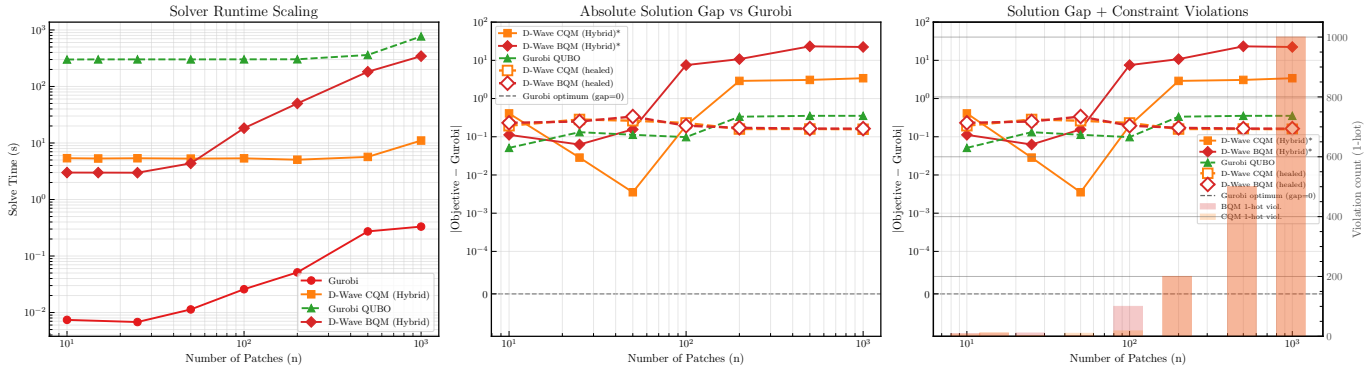

Fig. 1. **Study 1: Hybrid solver benchmark (10–1,000 patches).** *Left (Runtime):* Gurobi CQM scales 0.006–3.8 s; D-Wave CQM hybrid ≈5–11 s (QPU ≤1.3% of wall time); BQM hybrid 3–344 s; Gurobi BQM times out (≥300 s) at every scale. *Middle (Absolute Gap):* Solid = raw objectives; open/dashed = healed (section IV-D). CQM gap below 10^{-2} at n=25–50 after healing; raw inflation 2.9–3.4 at n≥200. Healed CQM gap ∼0.1–0.2 at scale; healed BQM gap >0.1 everywhere. *Right (Gap + Violations):* Violation count bars (secondary axis) show actual patches with $Y_{p,c}$>0.5 for more than one crop (post-processed, not the built-in validator). Both solvers accumulate violations at every scale: BQM 9–11 at n≤25, rising to 100 at n=100 and all n patches at n≥200; CQM 2–12 at n≤50, escalating similarly. The validator counter reports zero at n≥50, masking this systematic infeasibility. Gurobi CQM dominates on all metrics.

TABLE IV
QPU TIME VS. CLASSICAL EMBEDDING TIME FOR MULTILEVEL(10) DECOMPOSITION. EMBEDDING CONSTITUTES 76–86% OF TOTAL TIME; QPU REMAINS 1.2–3.1% ACROSS ALL SCALES.

Plots	Total (s)	QPU (s)	Embed (s)	QPU%	Embed%
10	13.5	0.42	10.2	3.1%	75.7%
50	128.6	1.51	110.2	1.2%	85.7%
100	175.1	2.83	144.6	1.6%	82.6%
200	388.7	5.48	315.3	1.4%	81.1%
500	839.1	13.44	677.8	1.6%	80.8%
1,000	1,632.7	26.83	1,308.9	1.6%	80.2%

rounds, n_{part} grows linearly with plot count, and t_{anneal}≈100 ms per partition including QPU access latency [44], [45]. QPU annealing therefore scales linearly and remains below 27 s at 1,000 plots. Classical MinorMiner embedding grows from 10.2 s to 1,309 s over the same range, constituting **76–86%** of total runtime at every scale.

Table V compares solution quality across all methods at the 1,000-farm scale. Direct QPU embedding fails entirely, confirming that decomposition is mandatory for practical problem sizes. Among the tested methods, Multilevel(10) and CQM-First both achieve zero constraint violations with a gap of 0.171 at 1,000 plots; Multilevel(10) completes in 1,633 s vs. 3,495 s for CQM-First. Coordinated decomposition reaches a lower raw gap (0.137) but accumulates 23 one-hot violations—and its healed objective drops to −0.038, indicating that the apparent objective advantage is substantially driven by infeasible multi-crop assignments.

Gurobi decomposed baselines. To separate decomposition violations from QPU approximation, we also solve each decomposed subproblem exactly with Gurobi (marked * in tables and plots) for all decomposition strategies (Multilevel(10), CQM-First, and Coordinated shown in fig. 2).

Multilevel(10)* and Coordinated* consistently incur exactly 5 food-group violations regardless of scale—a structural artefact of the decomposition breaking global coverage constraints. CQMFirst* incurs 3 food-group violations. At the Multilevel and Coordinated granularity, Gurobi recovers near-optimal objectives within each subproblem (healed gap $\sim 10^{-3}$), demonstrating that the decomposition itself preserves solution quality well. CQMFirst* achieves a lower raw objective (∼0.353 vs ∼0.429), reflecting that the U-master step pre-selects foods before patch assignment, producing a slightly different feasible solution. This comparison confirms that much of the gap in QPU results originates from the decomposition and embedding pipeline rather than from QPU sampling quality.

TABLE V
SOLUTION QUALITY AT THE 1,000-FARM SCALE. GAP = $|Z_{\text{METHOD}} - Z^{\star}_{\text{GUROBI}}|$. "(H)" ROWS SHOW HEALED OBJECTIVES; FOR METHODS WITH ZERO VIOLATIONS HEALED = RAW. GUROBI DECOMPOSED METHODS ARE MARKED *; CQMFIRST* HAS 3 FOOD-GROUP VIOLATIONS; MULTILEVEL(10)* AND COORDINATED* EACH INCUR 5.

Method	Obj.	Gap	Viol.	Time (s)
Gurobi GT (optimal)	0.429	0.000	0	0.3
Multilevel(10) *QPU*	0.258	0.171	0	1,632.7
Multilevel(10) *QPU* (h)	0.258	0.171	—	
CQM-First *QPU*	0.258	0.171	0	3,495.4
CQM-First *QPU* (h)	0.258	0.171	—	
Coordinated *QPU*	0.293	0.137	23	3,058.0
Coordinated *QPU* (h)	−0.038	0.467	—	
Gurobi [Multilevel(10)]*	0.430	0.001	5	0.2
Gurobi [Multilevel(10)]* (h)	0.428	0.001	—	
Gurobi [CQMFirst]*	0.353	0.076	3	<0.1
Gurobi [CQMFirst]* (h)	0.352	0.077	—	
Gurobi [Coordinated]*	0.430	0.001	5	<0.1
Gurobi [Coordinated]* (h)	0.428	0.001	—	

Figure 2 presents the full comprehensive benchmark.

Table VI breaks down constraint violations by decomposition strategy across all tested scales (full data for all

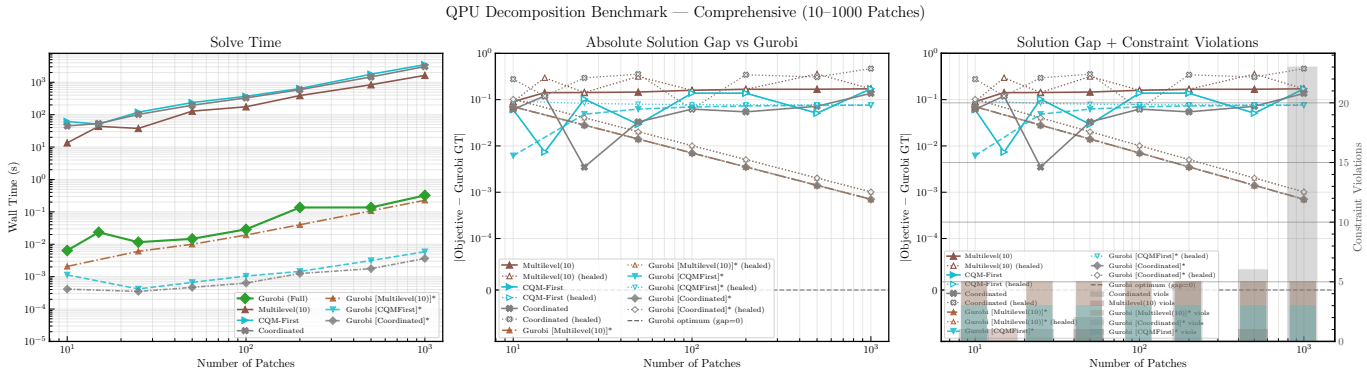

Fig. 2. **Study 2: QPU decomposition benchmark (10–1,000 plots).** Three full-span QPU methods with Gurobi decomposed baselines (dashed, *): Multilevel(10), CQM-First, Coordinated. *Left (Runtime):* Gurobi full-problem: 0.007–0.27 s. Multilevel(10): 14–1,633 s; CQM-First: 7–3,495 s (embedding 80+%). All Gurobi decomposed baselines complete in <10 s. *Middle (Absolute Gap):* Multilevel(10): 0.09–0.17, zero violations. CQM-First: 0.007–0.17, zero violations. Coordinated: raw 0.003–0.14 but 1–23 one-hot violations; healed gap rises substantially. Gurobi decomposed: Multilevel(10)* and Coordinated* recover gap $\sim 10^{-3}$ (healed). *Right (Gap + Violations):* Coordinated violations escalate 1→23 across scales, inflating its raw objective. Multilevel(10) and CQM-First maintain zero violations at all scales.

TABLE VI
CONSTRAINT VIOLATION SUMMARY BY DECOMPOSITION STRATEGY. COORDINATED IS THE ONLY QPU METHOD WITH ONE-HOT VIOLATIONS, ESCALATING TO 23 AT n=1,000 (38 TOTAL ACROSS ALL SCALES).

Method	Entries w/ Viol.	Total	Rate	Type
CQM-First	0/8 (0%)	0	None	—
Louvain	1/5 (20%)	1	Low	Food group
Multilevel(10)	3/8 (38%)	3	Med.	Food group
Multilevel(5)	3/5 (60%)	4	Med.	Food group
PlotBased	2/5 (40%)	2	Med.	Food group
Spectral(10)	3/5 (60%)	3	Med.	Food group
Coordinated	6/8 (75%)	38	High	One-hot

8 methods). Two violation patterns emerge: *food-group violations* arise when partition boundaries prevent global category constraints from being satisfied, and *one-hot violations* arise when coordination rounding assigns zero or multiple crops to a single plot.

D. Post-Hoc Feasibility Repair and Healed Objective

Several solvers return solutions that violate BP constraints, producing an infeasible raw objective Z_{raw}. We apply a greedy one-shot repair and report the *healed objective* $\tilde{Z} \leq Z_{\text{raw}}$ as a lower bound on the feasible quality attainable by that solver.

One-crop-per-patch violations ($\sum_c Y_{p,c} > 1$, both studies): for each violating patch p with conflicting crops $\mathcal{C}_p$, retain only the highest-benefit crop and subtract the minimum-benefit contribution:

$$\tilde{Z} = Z_{\text{raw}} - \sum_{p \in \mathcal{V}} \min_{c \in \mathcal{C}_p} b_c \cdot \frac{A_p}{A_{\text{tot}}}, \tag{5}$$

where $\mathcal{V}$ is the set of violating patches and b_c is the benefit coefficient of crop c. Each repaired patch is left with exactly one crop.

Food-group coverage violations (Study 2/Gurobi decomposed only): when group g has no assigned patch ($\Delta_g > 0$ patches short), reassign the Δ_g minimum-benefit active patches to a representative crop of g:

$$\tilde{Z} -= \Delta_g \cdot \min_{p \in \mathcal{P}_{\text{act}}} b_{c(p)} \cdot \frac{A_p}{A_{\text{tot}}}, \tag{6}$$

where $\mathcal{P}_{\text{act}}$ excludes patches already covering group g. A large gap $Z_{\text{raw}} - \tilde{Z}$ signals that the reported objective is substantially inflated by infeasible multi-crop assignments. In figures, raw objectives use solid lines; healed objectives use dotted lines of matching colour (omitted where the two coincide).

V. DISCUSSION & CONCLUSION

This paper reported two complementary experiments—a hybrid solver benchmark and a decomposition study—that characterise where computational effort concentrates when mapping a structured agricultural allocation problem onto quantum annealing hardware. Three engineering tradeoffs emerge.

Penalty encoding breaks classical solvers but not D-Wave. The BP formulation is tractable by both classical and hybrid solvers when submitted explicit constraints: Gurobi solves instances optimally in 0.007–0.27 s across 10–1,000 plots, while D-Wave's CQM hybrid solver returns solutions in ∼5–11 s. Converting those constraints into QUBO penalty terms renders Gurobi ineffective at every tested scale (300+ s timeouts with objective degradation to Z=0 above 500 patches), whereas D-Wave's BQM hybrid solver still completes in 3–344 s. This is a structural consequence of destroying algebraic constraint information, not a solver deficiency [7], [29]. QA's energy-minimisation paradigm is unaffected, providing practical motivation for the quantum workflow even where decomposition is necessary.

Hybrid solvers are predominantly classical. QPU time constitutes 0.3–1.3% of D-Wave CQM wall-clock time and 0.2–2.4% of BQM time; the remainder is classical orchestration [46]. More critically, both D-Wave solvers return infeasible solutions at *every* tested scale: healed-objective

analysis reveals CQM inflation of 0.26–3.6 and BQM inflation of 0.04–23.4 across all scales. After healing, the CQM gap to Gurobi is $\sim$0.1–0.2, and BQM exceeds 0.1 everywhere. Gurobi on the native formulation dominates on every metric—speed, optimality, and feasibility—underscoring that hybrid solvers, despite handling penalty encoding, do not yet offer a competitive advantage for this problem class. This, without knowledge on the classical hardware accessed by the hybrid solvers.

QPU annealing scales well; embedding does not. Annealing time grows linearly with partition count and stays below 27 s at 1,000 plots (1.6% of total time for Multilevel(10)). Classical MinorMiner embedding grows from 10 s to 1,309 s over the same range, constituting 76–86% of total runtime. Among the QPU decomposition strategies, Multilevel(10) and CQM-First achieve zero violations with gaps of 0.09–0.17 across all scales, completing in 1,633 s and 3,495 s respectively at n=1,000. Coordinated reaches lower raw gaps but at the cost of escalating one-hot violations (23 at n=1,000), making its raw objectives unreliable.

In conclusion, Gurobi operating on the native formulation remains strictly superior on all metrics—speed (0.27 s vs. 1,633 s at best for QPU), optimality (exact solution vs. 2–40% gap), and feasibility (zero violations vs. 0–23). The central engineering contribution of this study is the quantitative decomposition of the pipeline needed to solve the Food Optimization problem—at satisfactory scale for real world impact—with Quantum Annealing: QPU time itself scales well and produces good subproblem solutions, but the classical embedding overhead constitutes 76–86% of end-to-end runtime, and decomposition introduces 1–17% objective loss even before accounting for QPU approximation. The primary path to improved end-to-end performance therefore lies in the classical pipeline: higher-connectivity QPU topologies such as the forthcoming Zephyr architecture [47], topology-aware embedding algorithms, larger native subproblem capacity, and parallelised multi-QPU execution [44]. With continued improvements in connectivity, embedding efficiency, and decomposition quality, our results provide a reproducible benchmark and methodology for practitioners evaluating quantum annealing in agricultural and broader structured resource allocation domains—supporting a credible path from proof-of-concept experiments to practical decision-support tools for regional agricultural planning.

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References

[1] S. Mudare, J. Jing, D. Makowski, X. He, Z. Liang, Z. Sims, T. C. Wanger, D. Tilman, F. Zhang, and W.-F. Cong, "Crop rotations synergize yield, nutrition, and revenue: a meta-analysis," *Nature Communications*, vol. 16, no. 1, p. 9552, 2025.

[2] M. A. Ana Esteso and A. Ortiz, "Sustainable agri-food supply chain planning through multi-objective optimisation," *Journal of Decision Systems*, vol. 0, no. 0, pp. 1–25, 2023. [Online]. Available: https://doi.org/10.1080/12460125.2023.2180138

[3] B. B. Lin, "Resilience in agriculture through crop diversification: Adaptive management for environmental change," *BioScience*, vol. 61, no. 3, pp. 183–193, 2011.

[4] T. M. Bowles, M. Mooshammer, Y. Socolar, F. Calderón, M. A. Cavigelli *et al.*, "Long-term evidence shows that crop-rotation diversification increases agricultural resilience to adverse growing conditions in North America," *One Earth*, vol. 2, no. 3, pp. 284–293, 2020.

[5] P. R. Hobbs, K. Sayre, and R. Gupta, "The role of conservation agriculture in sustainable agriculture," *Philosophical Transactions of the Royal Society B*, vol. 363, no. 1491, pp. 543–555, 2008.

[6] J. Dury, N. Schaller, F. Garcia, A. Reynaud, and J.-E. Bergez, "Models to support cropping plan and crop rotation decisions. A review," *Agronomy for Sustainable Development*, vol. 32, no. 2, pp. 567–580, 2012.

[7] L. A. Wolsey, *Integer Programming*. New York: John Wiley & Sons, 1998.

[8] T. Albash and D. A. Lidar, "Adiabatic quantum computation," *Reviews of Modern Physics*, vol. 90, no. 1, p. 015002, 2018.

[9] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[10] O. Titiloye and A. Crispin, "Quantum annealing of the graph coloring problem," *Discrete Optimization*, vol. 8, no. 2, pp. 376–384, 2011.

[11] D. Venturelli, D. J. Marchand, and G. Rojo, "Quantum annealing implementation of job-shop scheduling," 2015.

[12] D. Venturelli and A. Kondratyev, "Reverse quantum annealing approach to portfolio optimization problems," *Quantum Machine Intelligence*, vol. 1, no. 1, pp. 17–30, 2019.

[13] S. Yarkoni, E. Raponi, T. Bäck, and S. Schmitt, "Quantum annealing for industry applications: Introduction and review," *Reports on Progress in Physics*, vol. 85, no. 10, p. 104001, 2022.

[14] D-Wave Systems Inc., "D-Wave Advantage system: An overview," D-Wave Technical Report 14-1049A-A, 2020, d-Wave Advantage QPU with 5000+ qubits in Pegasus topology. [Online]. Available: https://www.dwavesys.com/media/s3qbjp0s/14-1049a-a_the_d-wave_advantage_system_an_overview.pdf

[15] ——, *Pegasus Topology: Technical Documentation*, D-Wave Technical Documentation, 2022, documentation for the Pegasus graph topology (degree 15) used in D-Wave Advantage QPUs. [Online]. Available: https://docs.dwavequantum.com/en/latest/quantum_research/topologies.html

[16] F. Glover, G. Kochenberger, and Y. Du, "A tutorial on formulating and using qubo models," *arXiv preprint arXiv:1811.11538*, 2018, comprehensive tutorial on QUBO problem formulation and complexity.

[17] A. Lucas, "Ising formulations of many NP problems," *Frontiers in Physics*, vol. 2, p. 5, 2014.

[18] K. Boothby, P. Bunyk, J. Raymond, and A. Roy, "Next-generation topology of D-Wave quantum processors," *arXiv preprint arXiv:2003.00133*, 2020. [Online]. Available: https://arxiv.org/abs/2003.00133

[19] V. Choi, "Minor-embedding in adiabatic quantum computation," *Quantum Information Processing*, vol. 7, no. 5, pp. 193–209, 2008.

[20] J. Cai, W. G. Macready, and A. Roy, "A practical heuristic for finding graph minors," *arXiv preprint arXiv:1406.2741*, 2014. [Online]. Available: https://arxiv.org/abs/1406.2741

[21] M. S. Könz, W. Lechner, H. G. Katzgraber, and M. Troyer, "Embedding algorithms for quantum annealers with restricted connectivity," *Physical Review X Quantum*, vol. 2, no. 4, p. 040322, 2021.

[22] H. N. Djidjev, "Logical qubit implementation for quantum annealing: augmented Lagrangian approach," *Quantum Science and Technology*, vol. 8, no. 3, p. 035013, 2023.

[23] S. Zbinden, A. Bärtschi, H. Djidjev, and S. Eidenbenz, "Embedding algorithms for quantum annealers with Chimera and Pegasus connection topologies," in *High Performance Computing: 35th International Conference, ISC High Performance 2020, Frankfurt/Main, Germany, June 22–25, 2020, Proceedings 35*, ser. Lecture Notes in Computer Science, vol. 12151. Springer, 2020, pp. 187–206.

[24] G. Bass, M. Henderson, J. Heath, and J. Dulny III, "Optimizing the optimizer: decomposition techniques for quantum annealing," *Quantum Machine Intelligence*, vol. 3, no. 2, p. 39, 2021.

[25] M. Booth, S. P. Reinhardt, and A. Roy, "Partitioning optimization problems for hybrid classical/quantum execution," D-Wave Systems, Technical Report, 2017.

[26] B. Aspvall, M. F. Plass, and R. E. Tarjan, "A linear-time algorithm for testing the truth of certain quantified boolean formulas," *Information Processing Letters*, vol. 8, no. 3, pp. 121–123, 1979.

[27] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness (Series of Books in the Mathematical Sciences)*, first edition ed. W. H. Freeman, 1979. [Online]. Available: http://www.amazon.com/Computers-Intractability-NP-Completeness-Mathematical-Sciences/dp/0716710455

[28] C. H. Papadimitriou, *Computational Complexity*. Reading, Massachusetts: Addison-Wesley, 1994.

[29] F. Barahona, "On the computational complexity of Ising spin glass models," *Journal of Physics A: Mathematical and General*, vol. 15, no. 10, pp. 3241–3253, 1982.

[30] D-Wave Quantum Inc., "Hybrid solver — Leap cloud platform," https://www.dwavequantum.com/solutions-and-products/cloud-platform/hybrid-solver/, 2024, accessed: 2026-03-06.

[31] A. D. King, J. Raymond, T. Lanting, R. Harris, A. J. Berkley *et al.*, "Coherent quantum annealing in a programmable 2,000 qubit Ising chain," *Nature Physics*, vol. 18, pp. 1324–1328, 2022.

[32] W. Vinci, T. Albash, and D. A. Lidar, "Nested quantum annealing correction," *npj Quantum Information*, vol. 2, p. 16017, 2016.

[33] W. Vinci, T. Albash, G. Paz-Silva, I. Hen, and D. A. Lidar, "Quantum annealing correction with minor embedding," *Physical Review A*, vol. 92, no. 4, p. 042310, 2015.

[34] K. Boothby, P. Bunyk, J. Raymond, and A. Roy, "Next-generation topology of d-wave quantum processors," 2020.

[35] G. Karypis and V. Kumar, "A fast and high quality multilevel scheme for partitioning irregular graphs," *SIAM Journal on Scientific Computing*, vol. 20, no. 1, pp. 359–392, 1998.

[36] B. Hendrickson and R. Leland, "A multilevel algorithm for partitioning graphs," in *Proceedings of the 1995 ACM/IEEE Conference on Supercomputing*. ACM, 1995, p. 28.

[37] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, p. P10008, 2008.

[38] U. Von Luxburg, "A tutorial on spectral clustering," *Statistics and Computing*, vol. 17, no. 4, pp. 395–416, 2007.

[39] J. Shi and J. Malik, "Normalized cuts and image segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 8, pp. 888–905, 2000.

[40] A. Toselli and O. Widlund, *Domain Decomposition Methods–Algorithms and Theory*. Berlin: Springer, 2005.

[41] A. M. Geoffrion, "Generalized benders decomposition," *Journal of Optimization Theory and Applications*, vol. 10, no. 4, pp. 237–260, 1972.

[42] Z. Bian, F. Chudak, W. G. Macready, and G. Rose, "The ising model: Teaching an old problem new tricks," D-Wave Systems White Paper, 2020.

[43] P. Date, D. Arthur, and L. Pusey-Nazzaro, "Efficiently embedding qubo problems on adiabatic quantum computers," *Quantum Information Processing*, vol. 18, no. 4, p. 117, 2019.

[44] E. Pelofske, G. Hahn, and H. N. Djidjev, "Parallel quantum annealing," *Scientific Reports*, vol. 12, no. 1, p. 4499, 2022.

[45] J. Marshall, D. Venturelli, I. Hen, and E. G. Rieffel, "Power of pausing: Advancing understanding of thermalization in experimental quantum annealers," *Physical Review Applied*, vol. 11, no. 4, p. 044083, 2019.

[46] D-Wave Systems. General qpu solver properties. Section: "minimum-time-limit" (fragment: #minimum-time-limit). Ocean SDK version 9.2.0. [Online]. Available: https://docs.dwavequantum.com/en/latest/quantum_research/solver_properties_all.html#minimum-time-limit

[47] K. Boothby, C. Enderud, T. Lanting, R. Molavi, N. Tsai, M. H. Volkmann, F. Altomare *et al.*, "Architectural considerations in the design of a third-generation superconducting quantum annealing processor," *arXiv preprint arXiv:2108.02322*, 2021.